



Friction microdynamics in the time and frequency domains: Tutorial on frictional hysteresis and resonance in precision motion systems

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ABSTRACT

This paper provides an overview and tutorial on the key behaviors of mechanical friction in both the time and frequency domains to understand friction effects on precision motion control systems, and presents implications of this behavior for high-precision applications. Friction behaves differently depending on the operating regime. In the steady sliding regime, friction is a function of the relative velocity of two objects sliding against each other, and characterized by static, Coulomb and viscous terms. We observe more complex frictional behaviors as the velocity and displacement are near zero, which is the pre-sliding regime. Friction in this regime depends on the displacement of objects in contact, acting as a nonlinear hysteretic spring. In this paper, we focus on the time- and frequency-domain phenomena caused by the nonlinear hysteretic elasticity of such pre-sliding friction. Frictional microdynamic behaviors in each domain are experimentally studied with a servomotor system testbed and using the Generalized Maxwell-Slip (GMS) friction model. Friction model identification methods based on experimental observations are also discussed in both the time- and frequency-domains. We also present structures for accurate time-domain simulation of friction.

1. Introduction

The purpose of this paper is to 1) provide an overview and tutorial on friction dynamic behavior and its modeling methods, 2) show simulation issues on the commonly-used Coulomb model causing non-physical regenerative chatter errors, and 3) present experimental friction phenomena and their physical interpretation in both the time and frequency domains using a simple servomotor system and an easy-to-implement empirical friction model.

Friction can be understood in two distinct regimes: the sliding regime and the pre-sliding regime [1]. Frictional behavior in the sliding regime, also known as the gross-sliding regime, is characterized as a velocity-dependent drag force between two objects in relative motion. Stribeck [2,3] established a model of this sliding friction by encompassing the static, Coulomb, and viscous frictions. In the pre-sliding regime, on the other hand, friction no longer acts as a simple drag force, but has a nonlinear hysteretic dependency on the position displacement of objects in contact [4]. Friction is a particular challenge in such a microdynamic regime, and it is important to physically understand the frictional behaviors and learn how to counteract the consequences including limit cycles, bigger tracking errors, and large settling time [5,6].

This paper primarily focuses on the hysteretic behavior in the pre-

sliding regime. We experimentally explore friction behaviors in both the time-domain and frequency-domain with the representative phenomena of input-dependent hysteresis and varying frictional resonances. Based on the experimental observations, we also introduce a friction model parameter identification method in each domain, which can be a convenient tool to establish a friction model and also to improve performance in precision control applications by model-based friction compensation techniques [7–9].

Note that while our prior article [9] mainly discusses the frequency-domain identification method and its usage for friction compensation, this tutorial paper focuses more on 1) the friction microdynamic behavior in both the time- and frequency-domain, as well as the identification methods and 2) the physical interpretation of nonlinear friction characteristics including input-dependent hysteresis, input-dependent low-frequency transfer function magnitude and phase lag, and input-dependent varying frictional resonances. We also present in this paper the simulation issues of the commonly-used Coulomb friction model to demonstrate the importance of accurate friction simulation.

The outline of this paper is as follows. We start, in Section 2, with reviewing prior art in friction modeling to understand efforts to mathematically model the frictional behaviors in both the sliding and pre-sliding regimes. Among the introduced models, we adopt the Generalized Maxwell-Slip (GMS) model [10] to help understand the physics

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of the friction effects. This model is also used to introduce the time- and frequency-domain parameter identification methods. We compare, in Section 3, the simulated time responses of a servomotor model with bearing friction, modeled either by the Coulomb model or the GMS model, in order to show the importance of understanding friction microdynamic behavior in high-accuracy motion control systems. In Section 4, we present a rotary motor testbed used for the friction experiments. The time-domain behavior of friction is discussed in depth in Section 5, where we introduce a simple test to demonstrate the pre-sliding friction effects on the settling of a spinning motor shaft. In Section 6, we discuss in detail the frictional phenomena in the frequency domain, such as input-level-dependency, low frequency phase lag, and moving resonances. At the end of Sections 5 and 6, we introduce the time- and frequency-domain identification methods, respectively, to identify the parameters of the GMS friction model. Section 7 then concludes the paper with suggestions for future work.

2. Prior art

In this section, we present representative aspects of friction along with known empirical friction models in the literature which are relevant to our study in this paper.

2.1. Classic friction model

Friction in the sliding regime is well established over the last century as a velocity-dependent drag force. The simplest model of this friction force is known as the Coulomb friction model. Here $F_f = \mu F_N$, where μ is the friction coefficient and F_N is the normal force. This model is often augmented with static friction (stiction) and a linear viscous drag term. Stribeck [3] developed a classic model by introducing continuous velocity dependency in the sliding regime, in the form of

$$\begin{aligned} F_f(v) &= \text{sgn}(v)s(v) + \sigma_2 v \\ &= \text{sgn}(v) \left(F_c + (F_s - F_c) \exp \left| \frac{v}{v_s} \right|^\delta \right) + \sigma_2 v. \end{aligned} \quad (1)$$

This classic friction model is called the Stribeck curve as illustrated in Fig. 1. Here F_c , F_s , v_s , δ and σ_2 represent the Coulomb friction, static friction, Stribeck velocity, shape factor, and viscous friction coefficient, respectively. Note that the Stribeck curve $F_f(v)$ consists of a velocity-weakening curve $s(v)$ and a velocity-strengthening curve $\sigma_2 v$ [11].

The Stribeck curve is a function of the relative velocity of the sliding objects. Data required to identify this curve can be experimentally obtained by measuring friction force over a range of constant velocities. The model's biggest drawback is that it includes a discontinuity at zero velocity, which is the pre-sliding regime. This limitation means that the Stribeck model does not encompass dynamical phenomena such as friction lag [12], rate-dependent breakaway force [13,14], and hysteresis with non-local memory [4,15,16]. This shortcoming was partially addressed in Refs. [17,18] where Tan et al. developed an empirical friction model as a modified Langevin function which maintains

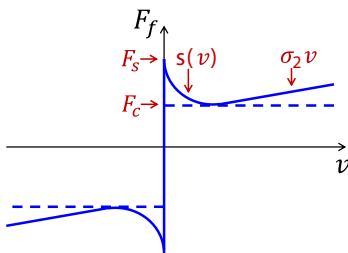


Fig. 1. Schematic of representative frictional behavior in the sliding regime as Stribeck curve with stiction F_s , Coulomb friction F_c , velocity-weakening curve $s(v)$, and velocity-strengthening viscous friction $\sigma_2 v$.

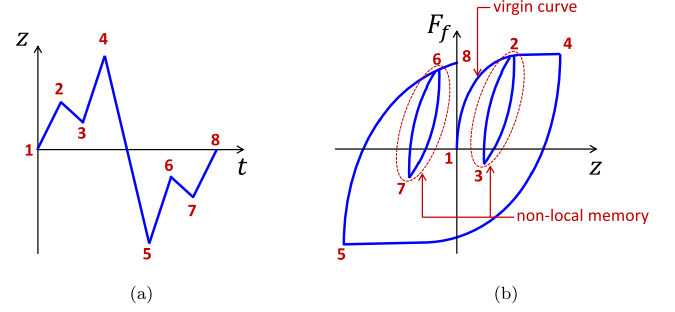


Fig. 2. Representative frictional behavior in the pre-sliding regime: (a) On the left, a small motion bidirectional position trajectory with velocity-reversals. (b) On the right, frictional hysteresis with non-local memory corresponding to the pre-sliding trajectory of (a).

continuity even at zero velocity. However, the hysteretic behavior of pre-sliding friction is not included in this model.

2.2. Modern friction models

A number of empirical models have been introduced in the literature attempting to represent the hysteretic elastic behavior of the pre-sliding regime as depicted in Fig. 2b in addition to the classic aspects of sliding friction.

Dahl [19,20] introduced a model describing the pre-sliding regime friction in the form of

$$\frac{dF_f(x)}{dx} = \sigma_0 \left| 1 - \frac{F_f}{F_c} \text{sgn}(v) \right|^n \text{sgn} \left(1 - \frac{F_f}{F_c} \text{sgn}(v) \right) \quad (2)$$

based on the hypothesis that the relationship of friction force $F_f(x)$ and displacement x resembles the stress-strain behavior of ductile materials. This model uses a shaping factor n to shape the elastic behavior of the pre-sliding friction. The stiffness parameter σ_0 represents an initial stiffness of the contact at the velocity reversal. This sets the slope of the friction-displacement curve for small displacements. These parameters, however, provide limited degrees of freedom to match experimental friction hysteresis, mostly resulting in a lower turnaround stiffness than actual friction. Instead of using such parameters for the analytical slope function, Wier and Alleyne [21] used an empirical slope function in their data-based approach. By linearly interpolating a measured friction slope and creating look-up tables, they improved the performance of the Dahl friction model to represent the frictional hysteretic elasticity more closely. Feldman et al. [22] also used a Dahl model in their force mapping technique to decompose the frictional resistance force into nonlinear elastic and damping forces so as to model the contact dynamic motion. We however note here that the Dahl model takes a simple approach to the sliding behavior as Coulomb friction without the stiction and the rate-dependency of the Stribeck curve.

Canudas de Wit et al. [23] introduced the LuGre model, where a state equation with the pre-sliding displacement z is used to model the friction in both regimes, including the Stribeck curve, as

$$\begin{cases} \frac{dz}{dt} = v \left(1 - \text{sgn}(v) \frac{\sigma_0 z}{s(v)} \right) \\ F_f = \sigma_0 z + \sigma_1 \frac{dz}{dt} + \sigma_2 v. \end{cases} \quad (3)$$

Here σ_1 represents a microscopic viscous friction coefficient. The state and friction equations of this model imply that the pre-sliding friction eventually approaches the Stribeck curve $F_f(v)$, as in Eqn. (1), at the steady state, where the time rate of the state z equals zero. This indicates a smooth transition from the pre-sliding to the sliding regime. Even though the LuGre model as a state equation allows simple implementation and describes both friction regimes, it was experimentally shown in Refs. [24,25] that this model resulted in worse tracking

performance than the Dahl model in a friction-compensated motion control system. It appears that the shaping factor n in the Dahl model is responsible for this result because this factor provides an additional degree of freedom to better shape the pre-sliding friction. This also indicates that the pre-sliding friction plays a more significant role than the sliding friction when it comes to high-precision performance. That is, when a moving system approaches a precision position, the friction forces in the pre-sliding regime dominate.

Note that both Dahl and LuGre models do not encompass *non-local memory* [26,27]. This term is used to describe the ability to form internal hysteresis loops and return to the initial hysteresis curve when there are multiple velocity-reversals in the displacement, as illustrated in Fig. 2. As the position trajectory ramps down from the position 2 to 3 and up to the position 4 crossing the previous position 2 in Fig. 2a, the hysteretic curve in the force-position plane closes an internal loop and comes back to its original path, crossing the starting point (marked as 2 in Fig. 2b) of the inner-loop. This non-local memory characteristic is an experimentally observed behavior of the hysteretic pre-sliding friction, but is not captured by either the Dahl or LuGre models. For examples and more details, see the results simulated in Refs. [16,25].

In order to properly encompass the observed non-local memory characteristic of friction, the LuGre model was modified by Swevers et al. [16] to develop the Leuven model. This model adds a hysteresis function $F_h(z)$ to represent the non-local memory via

$$\begin{cases} \frac{dz}{dt} = v \left(1 - \operatorname{sgn}(v) \left(\frac{F_h(z)}{s(v)} \right) \left| \frac{F_h(z)}{s(v)} \right|^n \right) \\ F_f = F_h(z) + \sigma_1 \frac{dz}{dt} + \sigma_2 v. \end{cases} \quad (4)$$

In the Leuven model, the hysteresis function $F_h(z)$ is constructed using a Maxwell-Slip (MS) model [28] of N parallel-connected elementary blocks, as shown in Fig. 3. Each of these MS blocks in isolation has an elastic behavior as shown in Fig. 4a. The MS blocks are connected to position z with a stiffness k_i , and with a Coulomb slip force limit $\alpha_i F_c$, where the variable F_c represents the Coulomb friction.

2.3. Generalized Maxwell-Slip model

To include static and viscous friction effects, Al-Bender et al. [29] imposed sliding dynamics as a weighted Stribeck velocity-weakening curve $\alpha_i s(v)$ on the slipping phase of each Maxwell-Slip block. This yields the Generalized Maxwell-Slip (GMS) model. The behavior of each GMS block is illustrated in Fig. 4b. In the GMS model, each elementary block is in a state of either *stick* or *slip*. This state determines the friction behavior via the rate-state law:

$$\begin{cases} \frac{dF_i}{dt} = k_i v & , \text{ if stick} \\ \frac{dF_i}{dt} = \operatorname{sgn}(v) C \left(\alpha_i - \frac{F_i}{s(v)} \right) & , \text{ if slip} \\ F_f = \sum_{i=1}^N F_i(t) + \sigma_2 v. \end{cases} \quad (5)$$

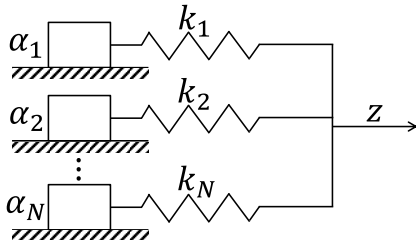


Fig. 3. Schematic of the Maxwell-slip model with a series of elementary blocks connected in parallel. Here z represents a small displacement in the pre-sliding regime. Each elementary block is characterized with stiffness k_i and normalized force saturation limit α_i at each of the slip blocks.

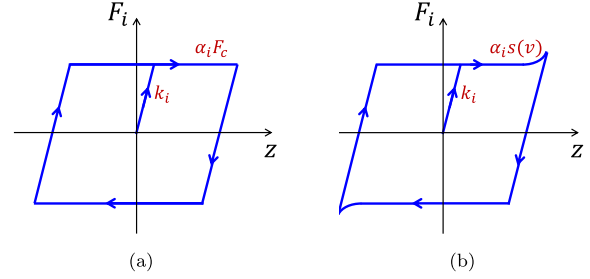


Fig. 4. Schematic behavior of each Maxwell-Slip elementary block with stiffness k_i and normalized saturation limit α_i : (a) Maxwell-Slip block with the Coulomb slip law, $\alpha_i F_c$. (b) Generalized Maxwell-Slip block with the slip dynamics, $\alpha_i s(v)$ representing velocity-dependence.

Here C represents an attraction factor determining how fast the slipping dynamics follows the velocity-weakening curve, $s(v)$. Note that the elementary block's behavior illustrated in Fig. 4b is exaggerated by using a large value of C . This allows illustrating the smooth transition from the pre-sliding to the gross-sliding regime. From the parallel structure, we obtain the total estimation of friction force F_f by summing the forces F_i from each block via Eqn. (5). The GMS model provides a relatively simple implementation for the purpose of real-time control. It has also been shown to be able to well-match experimentally-observed frictional phenomena. We have thus chosen to work with the GMS model to understand the time- and frequency-domain aspects of the friction in both the simulation and experiments in this tutorial paper. Note however that other dynamic friction models capturing the core characteristics of the pre-sliding friction can also be used to understand, model, and simulate the friction microdynamic behavior. For more details of such alternate models, see Refs. [21,28,30].

3. Friction simulation: significance of friction microdynamics

Accurate simulation of frictional behavior is important in understanding and controlling precision motion systems, and can be difficult because it has to encompass friction dynamics in both the sliding and pre-sliding regimes. The Coulomb model, for instance, is widely used to model and compensate friction in various servo-systems due to its simplicity. However, such a simple model can cause non-physical chattering behavior, thereby impairing servo performance when actually used to compensate friction. This is due to the absence of friction microdynamics in the Coulomb model.

In order to illustrate such issues and the importance of accurate friction simulation, we present in this section simulation comparisons of a servo-system with friction modeled by the Coulomb model and by the GMS model. A servomotor with ball-bearing friction shown in Section 4 is used for the simulation of spin-down free responses. Note that we choose a simple plant, which is a free mass of rotor, in order to 1) isolate friction from other possible system dynamics, hence no confounding effects and 2) emphasize the importance of friction microdynamics in modeling and compensating friction for servo applications. The simulation is programmed in LabVIEW and the relevant codes are available online [31].

The plant servomotor with ball-bearing friction is modeled as

$$J\ddot{\theta} = T_{\text{input}} - T_{\text{friction}} \quad (6)$$

where J , θ , T_{input} , and T_{friction} are respectively the rotor mass moment of inertia, rotational displacement, applied torque input to the motor model, and friction torque simulated by either the Coulomb model or the GMS model. The spin-down free response of this motor to a torque pulse is simulated. We input a torque pulse of 0.007 Nm at 20 ms for a duration of 50 ms to the motor model with the rotor moment of inertia of $J = 1.96 \times 10^{-5} \text{ kg}\cdot\text{m}^2$. The Coulomb model uses a Coulomb friction of 0.004 Nm, and the GMS model uses parameters identified by the

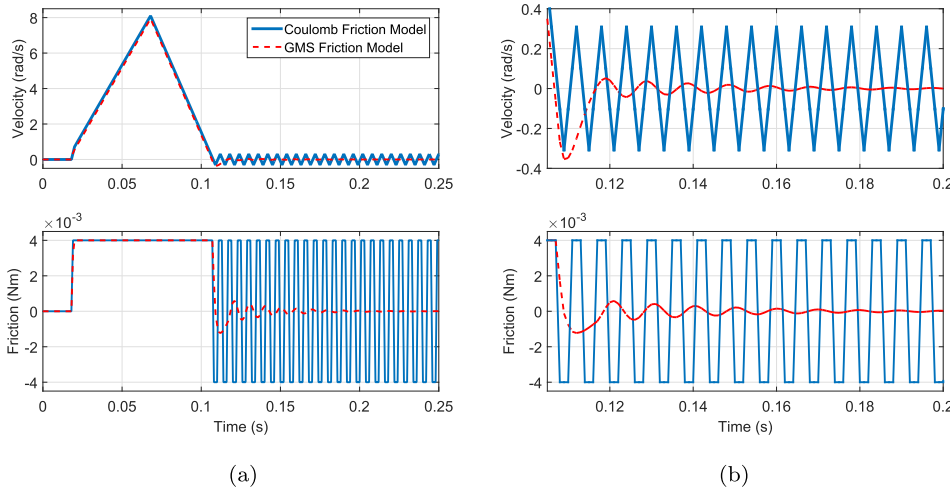


Fig. 5. Friction simulation comparison of the free response of a rotary motor to a torque pulse input: (a) On left, free responses of velocity and associated friction, compared between Coulomb friction model (blue solid line) and GMS friction model (red dashed line). (b) On right, magnified view of settling responses. Note non-physical self-excitation in steady-state associated with Coulomb model. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

time-domain method discussed in Section 5.3.

Fig. 5 compares the resultant motor free responses between the Coulomb model (blue solid line) and GMS model (red dashed line). Due to the discontinuity of the Coulomb model at zero velocity failing to include friction microdynamic behavior, the friction torque switches between the positive and negative Coulomb friction, thereby continuously regenerating velocity errors and limit-cycling in steady-state, which is clearly a non-physical behavior. With the GMS friction model, on the other hand, the motor velocity response settles down to zero as the friction torque decays as shown in Fig. 5b. The simulation result with the GMS friction model agrees with the experimental motor free responses presented in Section 5.1. Note that the ringing behavior at settling indicates the springiness of the pre-sliding friction, which causes frictional hysteresis and resonances. We discuss in detail such friction microdynamic phenomena in Sections 5 and 6.

4. Experimental setup

Our experimental setup shown in Fig. 6 uses a servomotor with a high-resolution rotary encoder to study frictional behaviors in both the time and frequency domains. The servomotor is a slotless and brushless Aerotech BMS 60. Using such motor 1) minimizes the extraneous nonlinear effects of brush friction (no brushes) and cogging (no slots in a surface-wound motor), and 2) allows us to study the friction caused by the ball bearings in the motor. Note that since the ball bearings are the friction source, the term *pre-rolling* might be technically correct rather than *pre-sliding*. We however keep using the more general term *pre-sliding* hereafter because pre-rolling friction essentially has the same characteristics as pre-sliding friction [32,33]. The rotary encoder installed in the motor provides an interpolated resolution of 10^6 counts per revolution. The position information is input to an FPGA encoder

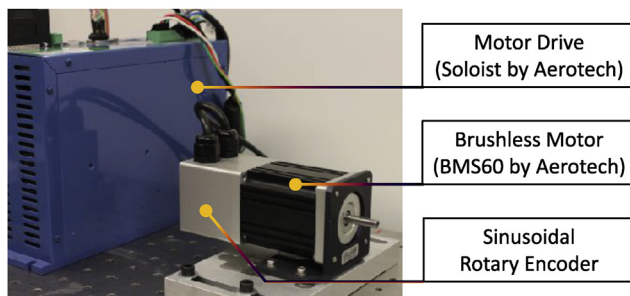


Fig. 6. Experimental setup consisting of a brushless and slotless servomotor (BMS60 by Aerotech), sinusoidal rotary encoder at the back of the motor, and motor drive (Soloist Hpe10 by Aerotech).

counter, which is implemented on a National Instruments (NI) PXI 8110 real-time controller.

5. Time domain behavior: hysteresis

The representative behavior of friction in the time domain is the hysteretic stiffness illustrated in Fig. 2b. We first introduce a simple experiment which directly illustrates the nonlinear friction elasticity using the measured data of servomotor's position and velocity. Next, we measure the frictional hysteresis in the torque-position plane and discuss its details. The description of a time-domain identification method follows with simulation results to show its fidelity.

5.1. Motor spin free response

A simple test can demonstrate the key behavior of pre-sliding friction. The shaft of the servomotor shown in Fig. 6 is spun by hand with the motor coils open-circuited, and then released. During the resulting spin-down to zero velocity, the position and velocity of the shaft are measured by the rotary encoder at a sampling rate of 2 kHz. The resulting free-spin responses of the motor shaft are plotted in Fig. 7.

The spin illustrated in Fig. 7 is roughly 0.55 rad in amplitude with a peak velocity of 7.6 rad/s. In this test, the friction of the motor ball bearing acts as a near-constant drag torque during continuous rotation, as evidenced by the near-constant slope of the decreasing velocity trajectory. However, when the velocity approaches zero, the friction shows elastic behavior, as seen via the oscillations in the magnified views. Note that the frequency of the oscillation is on the order of 50–60 Hz. Further, the resonant frequency increases with decreasing amplitude, which is consistent with the small-amplitude-stiffening nonlinear spring characteristic of the GMS friction model. This observation agrees with the nonlinear hysteretic behavior of pre-sliding friction [1,4] observed in the literature. It is also consistent with our measurements as presented in Section 5.2. The observed oscillation is also much lower in frequency than flexible modes of the motor structure; this is a rigid-body resonance of the motor inertia against the nonlinear spring of the pre-sliding bearing friction.

5.2. Frictional hysteresis

In order to obtain the hysteresis curves of the pre-sliding friction, the servomotor is operated in a position control loop. A series of sinusoidal position trajectories with varying amplitudes in the pre-sliding regime are commanded while measuring both the angular position of the servomotor using the encoder and the motor drive current command, which is assumed proportional to the applied torque for this

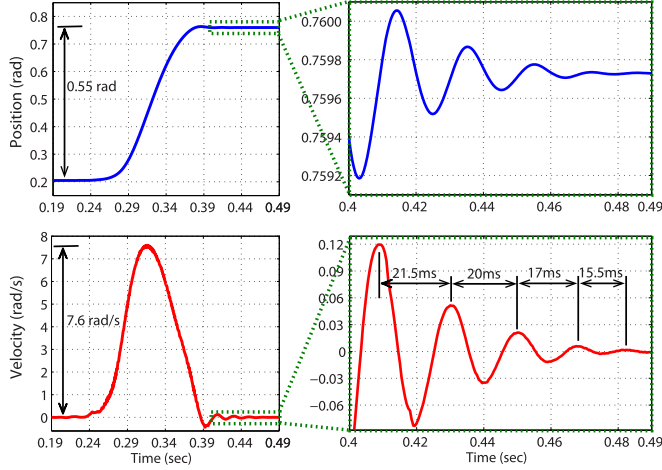


Fig. 7. Experimentally-measured free spin-down response of the motor shaft manually spun by hand. Position is measured in radians and velocity is measured in radians per second. Near the zero velocity point, the motor rotor inertia resonates against the friction elasticity. On the right, magnified responses in the settling region show the nonlinear elasticity of the pre-sliding friction with a shortening period of oscillation associated with higher incremental friction stiffness as the shaft slows to a stop.

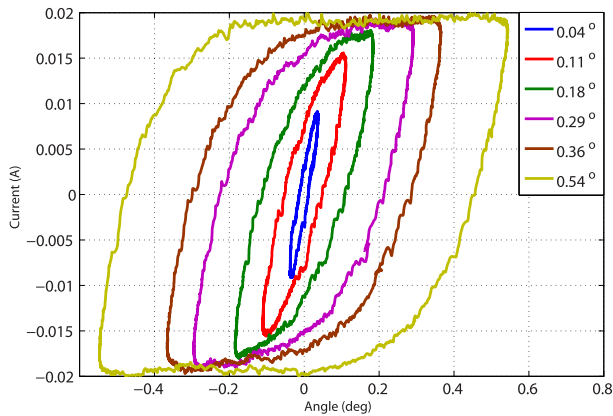


Fig. 8. Experimentally-measured current versus angle input-level-dependent hysteretic behavior of the servomotor pre-sliding friction.

Lorentz-force actuator. Motor commutation is supplied by the motor drive, so the current command is for the zero-to-peak amplitude of the three-phase currents. Note that the applied torque can be considered as the estimate of the friction torque because the inertial torque is relatively insignificant for the very low angular velocities associated with this small-angle motion. Frictional hysteresis curves are then plotted with the abscissa and ordinate being the pre-sliding angular position and corresponding current command, respectively. Typical resultant hysteresis curves are shown in Fig. 8.

The set of hysteresis curves in Fig. 8 illustrates that the hysteresis loop becomes wider as the input position amplitude increases, and the pre-sliding friction starts to saturate at approximately 0.3° , with a saturating current limit of 0.02 A. This result suggests that an angular displacement of 0.3° is the gross limit of the pre-sliding regime. Beyond this gross saturation point, all GMS blocks, which are representing asperities in mechanical contact, are in the slip phase, and converge to the Coulomb model. Note that a large enough input level needs to be commanded so that the measured hysteresis curve can contain all the hysteretic stiffness information of the pre-sliding friction. For instance, the innermost hysteresis loop in Fig. 8, with the input level below the gross saturation point, does not reach its maximum and only shows the

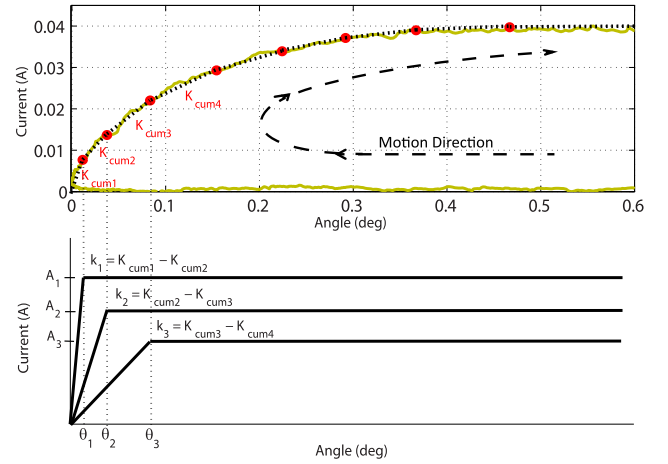


Fig. 9. Procedure of the hysteresis profile fitting identification method with an artificial virgin curve to identify N number of stiffnesses k_i 's and saturation limits α_i 's. Each dot indicates the angular location where each GMS block begins to slip, and K_{cum_i} is the accumulated stiffness, which is given by the local slope of the curve.

initial stiffness. It is therefore important to experimentally sweep enough range of input level so as to span the whole pre-sliding regime.

Note that the so-called *virgin curve* [1], which is defined as the early part of hysteresis curve before hitting the very first gross saturation point, is not shown in Fig. 8 because we obtain the curves using steady-state data. The virgin curve, however, can be estimated using one of the existing outer curves because the outer curves are the doubled-version of the virgin curve according to Masing's rule [1,34]. Section 5.3 discusses in detail how we can create an estimated virgin curve and utilize it to identify the friction model parameters.

5.3. Time domain identification

Friction model parameter identification in the time domain entails fitting a measured hysteresis curve with N parallel-connected GMS blocks [7,30]. This hysteresis profile fitting method begins with rendering an artificial virgin curve. To this end, we extract the left half of the outermost hysteresis loop from Fig. 8 and shift the curve so that the leftmost point is placed at the origin as shown in Fig. 9. The shifted curve now is the doubled-version of the original virgin curve and thus can be used for the profile fitting. Note that the doubled-version curve does not need to be scaled because what we want to extract from the curve is the stiffness information which is the ratio of the ordinate to the abscissa.

The artificial virgin curve is then piece-wise linearly fitted with N elementary blocks to identify $2N$ of the pre-sliding friction parameters, namely N stiffnesses and saturation limits for each block. The fitting procedure with eight GMS blocks ($N = 8$) is described in Fig. 9. At the start of the curve, all the GMS blocks are in the sticking state, meaning that all the elementary springs are engaged. The first accumulated stiffness K_{cum_1} is therefore the total sum of all N stiffnesses. When the first break point θ_1 occurs, one of the blocks reaches its slipping state, thereby reducing the accumulated stiffness to K_{cum_2} , whose relation can be written, for a general case, as

$$k_i = \begin{cases} K_{cum_i} - K_{cum_{i+1}}, & \text{if } i \neq N \\ K_{cum_i}, & \text{if } i = N \end{cases} \quad (7)$$

where $i = 1, 2, \dots, N$ and k_i represent the index of the GMS blocks and the i^{th} elementary stiffness respectively. Graphically speaking, we identify the elementary stiffnesses k_i 's by taking the difference of the adjacent slopes of the piece-wise linear segments.

The normalized saturation limits α_i 's are then calculated with the

Table 1
Identified GMS model parameters by the time-domain method.

i	k_i (A/deg)	α_i
1	0.4118	0.1169
2	0.0805	0.0749
3	0.0589	0.1248
4	0.0352	0.1381
5	0.0274	0.1556
6	0.0192	0.1423
7	0.0187	0.1738
8	0.0062	0.0736

values of previously obtained stiffnesses and the breaking points θ_i 's, using the relations written as

$$\begin{cases} A_i = k_i \times \theta_i \\ \alpha_i = \frac{A_i}{\sum_{j=1}^N A_j} \end{cases} \quad (8)$$

where A_i indicates a non-normalized saturating limit. Note that the total sum of the saturation limits should equal one in order to provide a smooth transition from the pre-sliding to the sliding regime when all the GMS blocks are in the slipping phase.

Table 1 lists the identified parameters for our motor friction model using this time-domain profile fitting method. The resultant hysteresis curves simulated with the identified GMS model are then shown in Fig. 10, overlapped with the measured data. The simulation results show good agreement with the experimental data, demonstrating the fidelity of the time-domain parameter identification method.

The identified GMS model can be used in various model-based control schemes to estimate and compensate the friction in high-precision motion systems. As an example, Jamaludin et al. [7] study quadrant glitch problems, which are large tracking errors caused at quadrant points during circular motions of a two-axis positioner. As another example, in Ref. [8] friction in a ballscrew-driven system is compensated to enhance tracking performance using an experimentally-identified GMS model within a model-based feedforward controller.

6. Frequency domain behavior: resonance

Distinct friction behaviors are also observed in the frequency domain, mainly in the form of input-dependent resonant frequencies associated with the nonlinear hysteretic elasticity. In this section, we first present measured frequency responses of the servo system setup with different excitation levels and study the friction effects on the

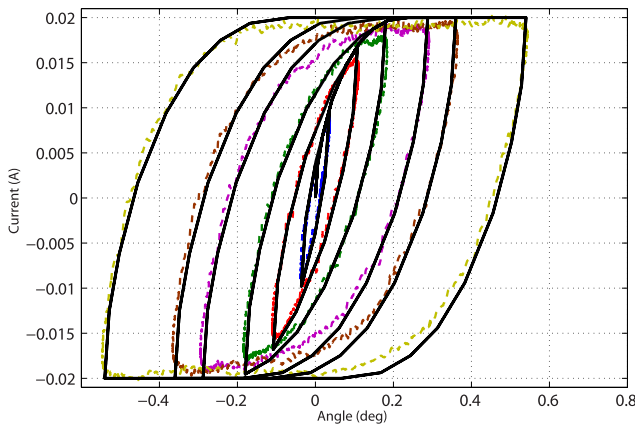


Fig. 10. Series of simulated hysteresis curves of servomotor current versus angle, in solid line, calculated from the identified GMS model by the time-domain curve fitting method. Simulated curves show reasonable agreement with the experimental data shown in dashed lines.

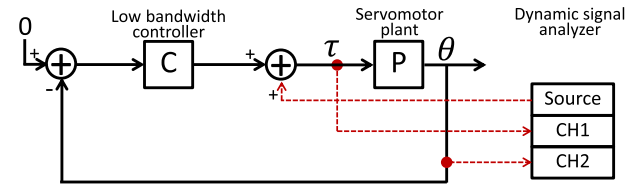


Fig. 11. Schematic configuration for frequency response measurements of the servo system. Low-bandwidth control loop with zero reference is used to prevent low frequency drift and potential bias during the measurement.

responses. Using the perspective gained from the experimental observations, we then introduce a frequency-domain resonance-based identification method for the GMS friction model.

6.1. Frequency response measurements

In order to correctly measure the plant frequency responses of the servomotor within the pre-sliding regime, we use a closed-loop configuration with a low-bandwidth controller as shown in Fig. 11.

Note that both the input torque τ and output position displacement θ need to stay within the pre-sliding regime during the measurements. For this purpose, the closed-loop approach with a low-bandwidth controller illustrated in Fig. 11 has several benefits. i) An open-loop configuration would cause a drift by integrating any DC offset at the input, and such drift can push the servomotor away from the pre-sliding regime. By closing a position loop, we prevent low frequency drift, thereby keeping the output position within pre-sliding. ii) A closed position loop, however, might introduce bias to the system responses when there are disturbances [35]. The low-bandwidth controller helps keep the possible bias small due to the ignorable control law in higher frequencies than the bandwidth, which is designed to be at around 1 Hz in our measuring loop. iii) The use of a low-bandwidth controller also helps keep the input torque level within the pre-sliding regime. By connecting the excitation source to the summing junction at the plant input, we can set the input torque level very closely to intended values, again due to the ignorably small control law in the frequency range of interest (10–100 Hz in our case) which is well above the designed loop bandwidth of 1 Hz. Note how a software-based dynamic signal analyzer is connected as shown in Fig. 11 to measure the plant frequency response.

Using our measurement loop, a series of frequency responses of the servomotor are measured and plotted in Fig. 12. Each curve represents a frequency response corresponding to a different input level of current

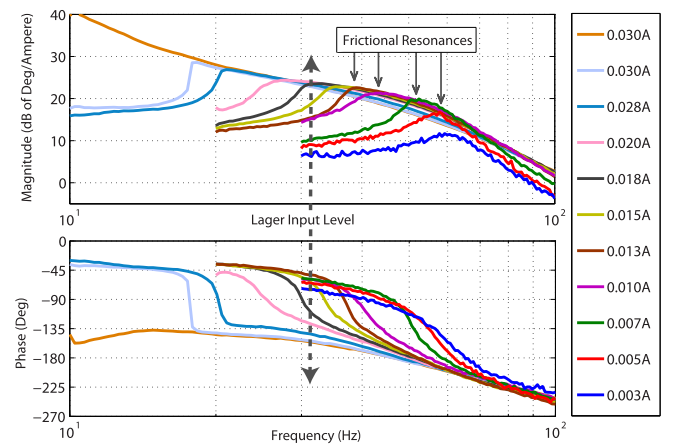


Fig. 12. Series of experimentally-measured frequency responses for varying excitation levels of servomotor for input current in Ampere to output angle in degrees. Data shows input level dependency and frequency-varying frictional resonant peaks. Magnitude plot is in decibel units of degrees per ampere, and the phase plot in degrees.

command, with the magnitude plot in the unit of degrees per ampere, and the phase plot in the unit of degrees.

From the measurements, we first notice how the responses are dependent upon the input amplitude level. The servomotor within the pre-sliding regime acts as a mass-spring system with a nonlinear spring, whose stiffness weakens as the input level increases. This behavior can be observed in the responses in the form of an increasing low-frequency transfer function magnitude, and a moving resonant peak toward lower frequency for a larger input level. This is because, in a mass-spring system, the resonant frequency is proportional to the square root of the stiffness, whereas the low-frequency value is inversely proportional to the stiffness. Such input-amplitude-dependent frequency responses of the hysteretic pre-sliding friction are also mathematically studied, by Helmick and Messner, in Ref. [36] using the describing function analysis associated with a Dahl model. There is, however, no experimental verification of these results in the paper, and the simulated resonances appear to differ in shape from typical experimental observations. In Section 6.2, we discuss in more depth the experimentally-observed input-dependent behaviors including the observed low frequency phase lag and amplitude-dependent resonant frequencies.

6.2. Low frequency phase lag and frictional resonance

Fig. 12 shows a phase lag of approximately -25° at low frequencies for the responses within the pre-sliding regime. This contrasts with a near-zero low-frequency phase lag of a standard mass-spring system. This additional phase delay stems from the nonlinear hysteresis of the pre-sliding friction stiffness, which dominates over the system inertia at low frequencies and causes a delay in displacement following the input torque. Fig. 13 demonstrates the input-level-dependent delay with a simulated hysteresis model. As illustrated, the delay of the displacement to the current (proportional to torque) decreases from t_2 to t_1 for the smaller input level. It is this time delay that causes the additional phase lag (displacement to torque) in the low frequency region. The phase lag would be near zero when the input amplitude is small enough to consider the frictional stiffness to be linear.

Note that in the experimental data of Fig. 12, there is a near-bifurcation in the data for 0.030 A current amplitude, with one response (orange line) approaching a free-mass curve, albeit with only about -140° phase shift, and one response (light blue) approaching a spring-dominated curve, albeit with about -25° phase shift. This bifurcated response implies that the current command of 0.030 A is on the edge of the breakaway point of the servomotor bearing friction. Given a torque

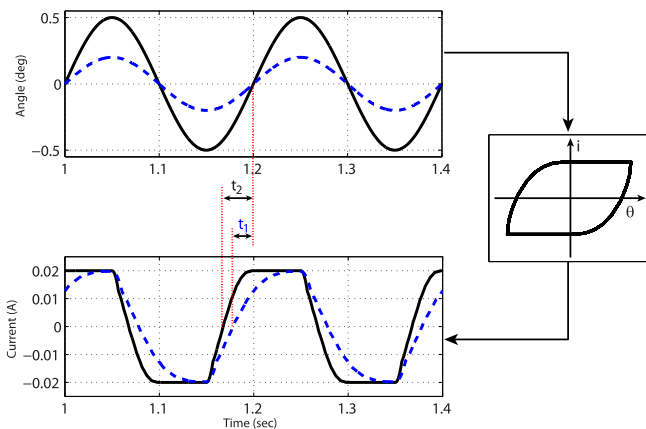


Fig. 13. Simulated demonstration of delay between torque (current) and displacement due to the frictional hysteresis in the pre-sliding regime for two motion amplitudes. Two different delays t_1 and t_2 indicate the input-level dependency. In this data, the motion is controlled to follow a sinusoid, and the simulation yields the required current drive profile. We assume an actuator in which torque is linearly proportional to current.

constant of 0.2 Nm/A according to the manufacturer, the static friction can then be estimated to be 6×10^{-3} Nm for our testbed.

We also observe another input-level-dependent phenomenon in the frequency domain, which is the moving resonant frequency with amplitude. Such frequency-domain behavior of friction has been observed in various electromechanical systems such as hard-disk drives [37–39], ultra-fast tool servos [40,41], and general servomotors [42–44].

It is important to understand this frequency-domain behavior of friction for the purpose of control design. The locations of the moving resonant peaks give a lower bound on the choice of loop crossover frequency in order to minimize the frictional resonance excitation as well as not to excite any potential higher-frequency eigenmodes of the system. This knowledge can be particularly beneficial when a controller is designed in the frequency domain using design techniques like loop-shaping.

The frictional resonant phenomenon is a consequence of the hysteretic friction spring, whose stiffness changes depending on input levels. The frictional spring, therefore, resonates against the system inertia at different frequencies according to the different input amplitudes as shown in Fig. 12. In such a sense, we can obtain the changing stiffness information directly from the frequency-domain measurements, namely the frictional resonances. This approach is elaborated in Section 6.3, and is also presented in Ref. [9].

6.3. Frequency domain identification

The necessary parameters of the GMS friction model can be identified in the frequency domain by using the frictional resonances. Each resonant peak location of each frequency response curve indicates that the system effectively experiences that amount of average stiffness against the system inertia, which leads us to the notion of an effective stiffness as

$$K_{eff_i} = \frac{(2\pi f_{R_i})^2 \times J}{K_t} \times \frac{\pi}{180} \text{ (A/deg)} \quad (9)$$

where f_{R_i} , J , and K_t are the frictional resonant frequency, mass moment of inertia of the rotor, and torque constant respectively. The values of J and K_t are given as 1.96×10^{-5} kg·m² and 0.2 Nm/A by manufacturer. Note that K_{eff_i} is scaled by the factor of $\pi/(180 \times K_t)$ to match the units to the measured data in amperes per degree.

The effective stiffness is considered as the representative stiffness that the servomotor experiences at each input amplitude, and is illustrated in Fig. 14. Note that I_i 's in the figure represent the varying amplitude levels of the commanded current input. The accumulated stiffnesses are then calculated as

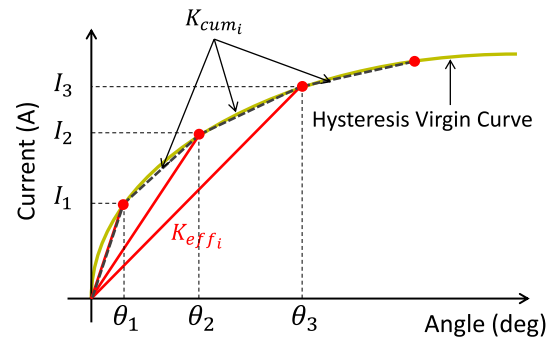


Fig. 14. Schematic description of parameter identification method using the effective stiffnesses of the pre-sliding friction K_{eff_i} obtained from the frictional resonances. Note the relation between K_{eff_i} and the accumulated stiffnesses K_{cum_i} , which is shown as a dashed line.

Table 2

Associated variables and identified parameters using the frequency-domain resonance-based method.

i	I_i (A)	f_{Ri} (Hz)	k_i (A/deg)	α_i
1	0.003	60.74	0.0508	0.0306
2	0.005	57.85	0.0693	0.0767
3	0.007	52.49	0.0603	0.1134
4	0.010	42.69	0.0030	0.0123
5	0.013	38.95	0.0266	0.1686
6	0.015	35.33	0.0059	0.0525
7	0.018	31.53	0.0164	0.2196
8	0.020	27.68	0.0169	0.3264

$$K_{cum_i} = \begin{cases} K_{eff_i} & \text{if } i = 1 \\ \frac{I_i - I_{i-1}}{\theta_i - \theta_{i-1}} = \frac{I_i}{K_{eff_i}} - \frac{I_{i-1}}{K_{eff_{i-1}}} & \text{if } i \neq 1. \end{cases} \quad (10)$$

After obtaining the accumulated stiffnesses as in Eqn. (10), the elementary stiffnesses k_i 's and normalized saturation limits α_i 's of the GMS friction model can be calculated in a similar fashion as for the time-domain identification method, via Eqn. (7) and Eqn. (8). The identified parameters are listed in Table 2. Using these parameters, we simulate frequency responses with the identified GMS friction model. The simulated response curves are shown in Fig. 15.

The simulation results in Fig. 15 show that the identified GMS model captures both the input-dependent low-frequency phase lags and the friction resonances, verifying the fidelity of the resonance-based identification method. Note that as discussed earlier in Section 6.2, the low-frequency phase lag approaches to near zero for small enough amplitudes, which is expected with a standard mass-spring system, as the input level becomes small enough (e.g. the case of 0.001 A in Fig. 15) for the frictional spring to behave linearly, i.e., to enter a regime of no slip in any of the GMS blocks.

We use the parameters identified by both the time- and frequency-domain methods to simulate frictional hysteresis curves, and compare them to an experimental hysteresis, as shown in Fig. 16. Note that the model fidelity is compared using the hysteresis curves in that 1) the identified frictional stiffness parameters cannot be directly compared since they are associated with different saturation limits, α_i 's and hence with different breaking angle points, θ_i 's and 2) the frequency-domain method also uses the concept of hysteretic virgin curve to identify the model parameters as depicted in Fig. 14.

From the figure, we see that the hysteresis curve simulated by the time-domain method closely follows the experimental curve. This is

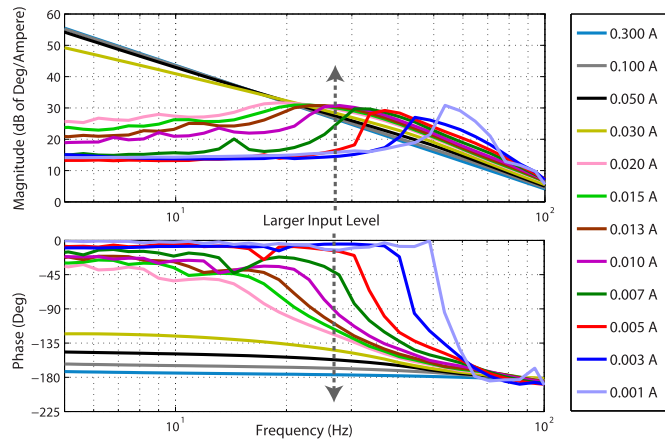


Fig. 15. Simulated amplitude-dependent frequency responses of the servomotor system with the GMS friction model identified by the frequency-domain resonance-based method. Input is the current in Ampere and the output is the motor angle in degrees.

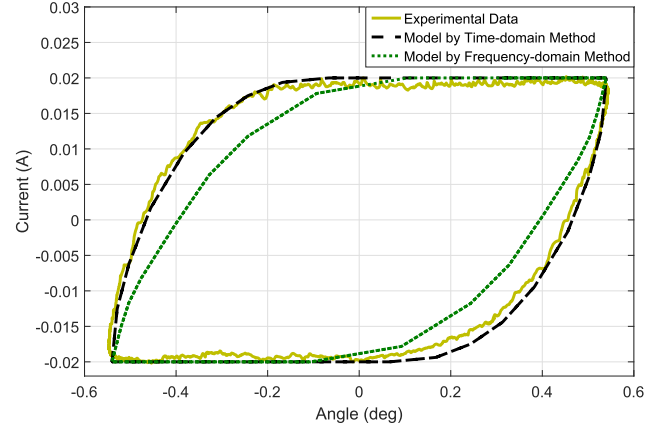


Fig. 16. Measured frictional hysteresis curve (solid line) compared to simulated models both by the time-domain method (dashed line) and the frequency-domain method (dotted line). Sinusoidal position reference with the amplitude of 0.54° is used for all cases.

because this method directly fits the measured hysteresis to identify the stiffness parameters as discussed in Section 5.3. The frequency-domain method, on the other hand, shows a small discrepancy. This is due to the fact that the resonance-based technique 1) uses the meta-data of frequency responses rather than the direct hysteresis measurements and 2) relies on the plant system characteristics such as the mass moment of inertia and the torque constant, as shown in Eqn. (9). Given that it is based on another data set, however, the hysteresis loop simulated by the frequency-domain method shows very reasonable agreement to the time-domain based curve. This implies that such identification method can be a useful tool for engineers and practitioners to understand and model systems with friction.

The resonance-based identification method can be also used together with control algorithms to mitigate the friction effects. For instance, the bearing friction of a servomotor system can be identified by frictional resonances and compensated with the GMS model in a feed-forward control algorithm such as in Ref. [9]. The reliability of the frequency-domain identification method is experimentally validated in the reference in both the frequency and time domains, as well as in the aspects of the closed-loop bandwidth and tracking performance.

7. Conclusions and future work

This paper studies frictional behaviors in the time and frequency domains with friction modeling and hardware experiments. The time-domain behavior of friction is represented by its nonlinear hysteretic elasticity. The GMS friction model helps to capture the hysteretic behavior in the torque-displacement plane by using parallel-connected elementary sliding blocks, each of which has its own stiffness and saturation limit. We experimentally demonstrate the bearing friction in a servomotor with a simple test of a free-spinning rotor coming to rest. The test shows that the frictional stiffness resonates against the rotor inertia, causing an oscillation when the shaft is settling to zero velocity. Given the continued widespread use of mechanical bearings in motion systems, it is important to understand such pre-sliding friction behaviors. Based on the experimental data on the frictional hysteresis, a time-domain parameter identification method of hysteresis profile fitting is also discussed.

We further study the behaviors of the pre-sliding friction in the frequency domain. The nonlinearity of friction is characterized by input-dependent phenomena in the system frequency responses: 1) low-frequency phase lag and 2) amplitude-dependent frictional resonance. Both behaviors are the results of the hysteretically changing frictional stiffness. In particular, knowledge of the locations of the resonant peaks helps in designing a proper controller so as not to excite those frictional

resonances and to make closed loop performance more independent of friction. We also present a resonance-based frequency-domain identification method for the GMS friction model and show how the identified model can capture the representative frequency-domain behaviors.

Understanding the behaviors of friction and how it affects a system in both the time and frequency domains can be a useful tool to understand friction effects. Both approaches to identify friction model parameters can be used with various control algorithms to improve system performance.

As future work, we suggest that more research is required to allow up-front prediction of frictional hysteretic elasticity directly from accurate mathematic models of mechanical elements in contact, such as bearings and transmission gears. The direct prediction of friction in both the sliding and pre-sliding regimes from first principles would allow the use of such estimations in the system design phase to save time and effort, and to achieve high performance in precision control systems. To our understanding, such a predictive friction model does not yet exist, but would surely be useful to enable more deterministic precision motion system design in the future.

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References

- [1] Al-Bender F. Fundamentals of friction modeling. *Proceedings, ASPE spring topical meeting on control of precision systems*. MIT; 2010. p. 117–22. April 11–13, 2010.
- [2] De Moerloose K, Al-Bender F, Van Brussel H. A generalised asperity-based friction model. *Tribol Lett* 2010;40(1):113–30.
- [3] Stribeck R. Die wesentlichen eigenschaften der gleit-und rollenlager-the key qualities of sliding and roller bearings. *Z Vereines Seutscher Ingenieure* 1902;46(38, 39):1342–8. 1432–1437.
- [4] Courtney-Pratt J, Eisner E. The effect of a tangential force on the contact of metallic bodies. *Proc Roy Soc Lond Ser A Math Phys Sci* 1957;238(1215):529–50.
- [5] Bucci BA, Cole DG, Ludwick SJ, Viperman JS. Nonlinear control algorithm for improving settling time in systems with friction. *IEEE Trans Contr Syst Technol* 2013;21(4):1365–73.
- [6] Dong X, Yoon D, Okwudire CE. A novel approach for mitigating the effects of pre-rolling/pre-sliding friction on the settling time of rolling bearing nanopositioning stages using high frequency vibration. *Precis Eng* 2017;47:375–88.
- [7] Jamaludin Z, Van Brussel H, Swevers J. Quadrant glitch compensation using friction model-based feedforward and an inverse-model-based disturbance observer. *Advanced motion control*, 2008. AMC'08. 10th IEEE international workshop on. IEEE; 2008. p. 212–7.
- [8] Yoon JY, Trumper DL. Friction modeling, identification, and compensation in a ballscrew driven system. *Proceedings, ASPE spring topical meeting on precision control for advanced manufacturing systems*. MIT; 2013. p. 33–8. April 21–23, 2013.
- [9] Yoon JY, Trumper DL. Friction modeling, identification, and compensation based on friction hysteresis and Dahl resonance. *Mechatronics* 2014;24(6):734–41.
- [10] Lampaert V, Al-Bender F, Swevers J. A generalized Maxwell-slip friction model appropriate for control purposes. *Physics and control*, 2003. *Proceedings*. 2003 international conference, vol. 4. IEEE; 2003. p. 1170–7.
- [11] Olsson H, Åström KJ, Canudas de Wit C, Gäfvert M, Lischinsky P. Friction models and friction compensation. *Eur J Contr* 1998;4:176–95.
- [12] Hess D, Soom A. Friction at a lubricated line contact operating at oscillating sliding velocities. *J Tribol* 1990;112(1):147–52.
- [13] Johannes V, Green M, Brockley C. The role of the rate of application of the tangential force in determining the static friction coefficient. *Wear* 1973;24(3):381–5.
- [14] Richardson R, Nolle H. Surface friction under time-dependent loads. *Wear* 1976;37(1):87–101.
- [15] Futami S, Furutani A, Yoshida S. Nanometer positioning and its micro-dynamics. *Nanotechnology* 1990;1(1):31.
- [16] Swevers J, Al-Bender F, Ganseman CG, Projogo T. An integrated friction model structure with improved presliding behavior for accurate friction compensation. *IEEE Trans Automat Contr* 2000;45(4):675–86.
- [17] Tan X, Modafe A, Ghodssi R. An empirical model for dynamic friction in micro-fabricated linear microball bearings. *American control conference*, 2005. *Proceedings of the 2005. IEEE*; 2005. p. 2463–8.
- [18] Tan X, Modafe A, Ghodssi R. Measurement and modeling of dynamic rolling friction in linear microball bearings. *J Dyn Syst Meas Contr* 2006;128(4):891–8.
- [19] Dahl P. Solid friction damping of spacecraft oscillations. *Guidance and control conference*. 1975. p. 1104.
- [20] Dahl PR. A solid friction model. *Tech. rep. Defense Technical Information Center (DTIC) Document*; 1968.
- [21] Weir NA, Alleyne AG. An improved dynamic friction model using a data-based approach. *ASME 2017 dynamic systems and control conference*. American Society of Mechanical Engineers; 2017. pp. V003T27A009–V003T27A009.
- [22] Feldman M, Zimmerman Y, Gissin M, Bucher I. Identification and modeling of contact dynamics of precise direct drive stages. *J Dyn Syst Meas Contr* 2016;138(7):071001.
- [23] Canudas de Wit C, Olsson H, Åström KJ, Lischinsky P. A new model for control of systems with friction. *IEEE Trans Automat Contr* 1995;40(3):419–25.
- [24] Lampaert V, Swevers J, Al-Bender F. Experimental comparison of different friction models for accurate low-velocity tracking. *Proceedings of the 10th mediterranean conference on control and automation (MED02)*. 2002.
- [25] Lampaert V, Swevers J, Al-Bender F. Comparison of model and non-model based friction compensation techniques in the neighbourhood of pre-sliding friction. *American control conference*, 2004. *Proceedings of the 2004, vol. 2. IEEE*; 2004. p. 1121–6.
- [26] Bertotti G. Hysteresis in magnetism: for physicists, materials scientists, and engineers. Academic Press; 1998.
- [27] Mayergoyz ID. Mathematical models of hysteresis. *IEEE Trans Magn* 1986;22(5):603–8.
- [28] Lampaert V, Swevers J, Al-Bender F. Modification of the Leuven integrated friction model structure. *IEEE Trans Automat Contr* 2002;47(4):683–7.
- [29] Al-Bender F, Lampaert V, Swevers J. The generalized Maxwell-slip model: a novel model for friction simulation and compensation. *IEEE Trans Automat Contr* 2005;50(11):1883–7.
- [30] Fukada S, Fang B, Shigeno A. Experimental analysis and simulation of nonlinear microscopic behavior of ball screw mechanism for ultra-precision positioning. *Precis Eng* 2011;35(4):650–68.
- [31] Yoon JY, Trumper DL. Friction micro-dynamics simulation codes. *DSPACE, Massachusetts Institute of Technology*; 2018<http://hdl.handle.net/1721.1/113847>.
- [32] Al-Bender F, Symens W. Characterisation of pre-sliding and pre-rolling friction. *World tribology congress III*. American Society of Mechanical Engineers; 2005. p. 297–8.
- [33] Al-Bender F, Symens W. Characterization of frictional hysteresis in ball-bearing guideways. *Wear* 2005;258(11–12):1630–42.
- [34] Lemaitre J, Chaboche J-L. *Mechanics of solid materials*. Cambridge University Press; 1990.
- [35] Söderström T, Stoica P. *System identification*. Prentice-Hall, Inc.; 1988.
- [36] Helmick D, Messner W. Describing function analysis of dahl model friction. *American control conference*, 2009. ACC'09. IEEE; 2009. p. 814–9.
- [37] Lai CY, Lewis FL, Venkataramanan V, Ren X, Ge SS, Liew T. Disturbance and friction compensations in hard disk drives using neural networks. *IEEE Trans Ind Electron* 2010;57(2):784–92.
- [38] Yoon JY. Design and testing of a high accuracy robotic single-cell manipulator Master's thesis Massachusetts Institute of Technology; 2011.
- [39] Abramovitch D, Wang F, Franklin G. Disk drive pivot nonlinearity modeling. i. frequency domain. *American control conference*, 1994, vol. 3. IEEE; 1994. p. 2600–3.
- [40] Ludwick SJ, Chargin DA, Calzaretta JA, Trumper DL. Design of a rotary fast tool servo for ophthalmic lens fabrication. *Precis Eng* 1999;23(4):253–9.
- [41] Lu X. Electromagnetically-driven ultra-fast tool servos for diamond turning PhD. thesis Massachusetts Institute of Technology; 2005.
- [42] Otsuka J, Masuda T. The influence of nonlinear spring behavior of rolling elements on ultraprecision positioning control systems. *Nanotechnology* 1998;9(2):85.
- [43] Hensen RH, van de Molengraft MJ, Steinbuch M. Frequency domain identification of dynamic friction model parameters. *IEEE Trans Contr Syst Technol* 2002;10(2):191–6.
- [44] Bucci BA, Viperman JS, Cole DG, Ludwick SJ. Evaluation of a servo settling algorithm. *Precis Eng* 2013;37(1):10–22.